

Matteo Brucato

*Helping humans
make better decisions.*

**Co-Founder &
Chief Scientific Officer**
OSM Data

osm-data.com

CONTACT

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EDUCATION

Ph.D. in Computer Science
UMass Amherst

Amherst, MA, USA · 2013 – 2021

Thesis: Package Queries —
Enabling Declarative and Scalable
Prescriptive Analytics in Relational
Data. Co-advisors: Peter J. Haas,
Alexandra Meliou, and Azza
Abouzied (NYU)

M.S. in Computer Science
University of Bologna

Bologna, Italy · 2011 – 2013

Summa cum laude. Thesis:
Temporal Information Retrieval.
Advisor: Danilo Montesi

B.S. in Computer Science
University of Bologna

Bologna, Italy · 2008 – 2011

Summa cum laude. Thesis: Mobile
systems for patient-reported
outcomes. Advisor: Danilo Montesi

TECHNICAL EXPERTISE

- Prescriptive analytics & in-
database optimization

SUMMARY

Co-founder and Chief Scientific Officer of OSM Data, where our team is building the first **Decisionhouse** — making prescriptive analytics and decision-making native to the data stack. At its core is **DeQL**, a SQL-like language that turns data into optimal decisions. Previously a researcher at Microsoft Research; PhD from UMass Amherst, pioneering the in-database optimization behind OSM — recognized with a CACM Research Highlight.

EXPERIENCE

Co-Founder & Chief Scientific Officer

Sept 2025 – present

OSM Data · Seattle, WA, USA

- Building the first Decisionhouse — a new category of data system in which decision-making is a first-class primitive alongside storage and analytics (VLDB '26 vision).
- Designing DeQL, a declarative SQL extension for prescriptive queries, and DeQL Studio, its interactive interface — which auto-selects the solver formulation to reach the same optimum **~30× faster** than a hand-written Gurobi MILP (VLDB '26 demo).
- Leading our founding team across engineering, research, and design.

Researcher

Oct 2021 – July 2025

Microsoft Research · Redmond, WA, USA

MLO (Machine Learning & Optimization) Group · Data Systems Group

- Shipped an improvement to Microsoft Azure's VM Allocator — a 1% packing-density gain and reduced CPU stranding, for **~\$100M/yr** projected savings.
- Cut Microsoft SQL Server index-tuning time **24×** (Wred, SIGMOD '24) via query-rewriting that preserves tuning accuracy; patent filed.
- Lifted LLM-to-optimization accuracy **10%** with OptiGuide; demoed to Microsoft Azure CVPs.
- Drove research-to-product transfer across teams, taking prototypes to production.

Research Assistant

Sep 2013 – May 2021

DREAM Lab, University of Massachusetts · Amherst, MA, USA

- Created Package Builder, a system that extends query engines to support the generation of *packages* — collections of tuples with global properties, computed as integer linear programs.
- Technical papers at VLDB 2024, SIGMOD 2020, VLDB 2016; demo papers at VLDB 2020 and VLDB 2014.
- Invited journal papers at CACM Research Highlights 2019, VLDB Journal 2018, SIGMOD Records 2017.

SELECTED PUBLICATIONS

Decisionhouse: Prescriptive Analytics in the Data Stack

Matteo Brucato, Fjodor Kholodkov, Soren Little, Jakob Mayer, Duc Nguyen

VLDB 2026

DeQL Studio: Declarative Decision-Making over Relational Data

Matteo Brucato, Fjodor Kholodkov, Soren Little, Jakob Mayer, Duc Nguyen

VLDB DEMO 2026

DeQL: A Decision Query Language for Prescriptive Analytics over Relational Data

Matteo Brucato, Fjodor Kholodkov, Soren Little, Jakob Mayer, Duc Nguyen

ARXIV 2026

Stochastic SketchRefine: Scaling In-Database Decision-Making under Uncertainty to Millions of Tuples

Riddho Ridwanul Haque, Anh Mai, Matteo Brucato, Azza Abouzied, Peter Haas, Alexandra Meliou

VLDB 2025

- Database systems & query engines
- LLM-to-optimization systems (NL→solver)
- Research-to-product technical leadership

OPEN SOURCE

github.com/osm-data

Reproducibility artifacts for our published research.

Wred: Workload Reduction for Scalable Index Tuning

Matteo Brucato, Tarique Siddiqui, Wentao Wu, Vivek Narasayya, Surajit Chaudhuri

SIGMOD 2024

Scaling Package Queries to a Billion Tuples via Hierarchical Partitioning and Customized Optimization

Anh L. Mai, Pengyu Wang, Azza Abouzied, Matteo Brucato, Peter J. Haas, Alexandra Meliou

VLDB 2024

sPaQLTools: A Stochastic Package Query Interface for Scalable Constrained Optimization

Matteo Brucato, Miro Mannino, Azza Abouzied, Peter J. Haas, Alexandra Meliou

VLDB DEMO 2020

Stochastic Package Queries in Probabilistic Databases

Matteo Brucato, Nishant Yadav, Azza Abouzied, Peter J. Haas, Alexandra Meliou

SIGMOD 2020

Scalable Computation of High-Order Optimization Queries

Matteo Brucato, Azza Abouzied, Alexandra Meliou

CACM 2019

Scalable Package Queries in Relational Database Systems

Matteo Brucato, Juan Felipe Beltran, Azza Abouzied, Alexandra Meliou

VLDB 2016

SELECTED TALKS

Presenting Decisionhouse (vision) & DeQL Studio (demo)

VLDB 2026

Enabling In-Database Decision Support with Package Queries

Gray Systems Lab, Microsoft · 2024

Package Queries: Scalable Prescriptive Analytics Close to the Data

Georgia Tech · Carnegie Mellon University · 2021

Package Queries: Enabling Declarative and Scalable Prescriptive Analytics in Relational Data

Harvard University, DASlab · MIT CSAIL, Data Systems Group · 2020–2021

Stochastic Package Queries for In-Database Optimization Under Uncertainty

INFORMS 2020

Scalable Package Queries in Relational Database Systems

Microsoft Research · 2017

Package Queries: From Tuples to Packages

AMPLab, UC Berkeley · 2015

SERVICE

PROGRAM COMMITTEE SIGMOD, VLDB & ICDE (2025–2027)

HONORS & AWARDS

2025 Distinguished Reviewer, SIGMOD 2025 — Berlin, Germany

2020 Best Demo Award, VLDB 2020 (sPaQLTools) — Tokyo, Japan

2020 Best Demo Runner-up Award, VLDB 2020 (SuDocu) — Tokyo, Japan

2019 Krithi Ramamritham Computer Science Scholarship — UMass Amherst

2018 CACM Research Highlight Award

2017 ACM SIGMOD Research Highlight Award

2017 Invited journal paper, VLDB Journal — Special Issue on Best Papers of VLDB 2016

2016 Best Papers of VLDB 2016 — New Delhi, India

2013 M.S. with Honors (summa cum laude), University of Bologna

2013 Best Young Researcher Award Nominee, RANLP 2013 — Hissar, Bulgaria

2013 Merit Scholarship — awarded to 0.2% of all students of the University of Bologna

2012 Research Scholarship, University of Bologna

2011 B.S. with Honors (summa cum laude), University of Bologna